(c) Briefly describe the overall operation of SSL Record Protocol with SSL Record Format. 7

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(d) Lists the alerts that are always fatal along with remainder of the alert, which are mentioned in the Alert Protocol

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B. E. (Eighth Semester) Examination, April-May, 2016

(New Scheme)

(ET & T Engg. Branch)

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CRYPTOGRAPHY & SECURE COMMUNICATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all the five questions. Part (a) of each question carry 2 marks. Attempt any two parts from (b), (c) and (d) each carries 7 marks.

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1. (a) Define a finite group and order of group with example. 2

(b) Briefly describe fast Exponentiation method with algorithm. Show the process for calculating $y = a^x$

using the algorithm in this case $x = 22 = (10110)_2$ in binary. The exponent has five binary bits. 7

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(c) Write Euclidean Algorithm to obtain the greatest common divisor and Extended Euclidean Algorithm to obtain the multiplicative inverse with example. 7

(d) State Euler's theorem : First version and second version. Find the result of :

(i) 6^{24} mod 35

(ii) 20^{62} mod 77 7

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2. (a) Write the differences between Block Cipher and Stream Cipher. 2

(b) Explain playfair cipher with proper steps of encrypting a message. Encrypt the message "COME SOON" with the keyword "GUIDANCE". 7

(c) Describe RSA Algorithm Encrypt the plain text 10, if $P = 47$, $Q = 17$ and $E = 5$ 7

(d) Explain Diffie-Hellman Key Exchange Algorithm and show how the algorithm is insecure against a Man-in-the-middle attack. 7

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3. (a) Write the functions that may be used to produce an authenticator. 2

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(b) Elaborate the Basic Uses of Message Authentication Code (MAC). 7

(c) Briefly describe the MD5 Algorithm, with the working steps in it. 7

(d) Draw the block diagrams of two approaches to Digital Signatures and explain DSA signing and verifying. 7

4. (a) What services are provided by IPSec? 2

(b) Describe the IPSec Architecture with IPSec Document Overview. 7

(c) Mention, how the most significant types of viruses can be categorized? 7

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(d) Illustrate three common firewall configurations with their block diagrams. 7

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5. (a) What is the purpose of dual signature. 2

(b) Explain, how SET specification lists the business requirements for secure payment processing and the features to meet the requirements. 7